EE5770 Course Project
5G NR PDSCH Chain with OFDM over an L-Tap Fading Channel

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May 6, 2026

Abstract

This report details the software implementation and evaluation of a complete 5G Physical Downlink Shared Channel (PDSCH) transceiver. Performance is evaluated over an L-tap frequency-selective Rayleigh fading channel. Two coding schemes—5G NR LDPC codes and zero-tail terminated convolutional codes—are compared under a strict matched-complexity constraint, alongside an evaluation of Zero-Forcing (ZF) and Minimum Mean-Square Error (MMSE) equalizers.

Contents

1 Introduction

The Physical Downlink Shared Channel (PDSCH) is the primary data-bearing channel in 5G NR. This project investigates the error-rate performance and decoding complexity of LDPC versus convolutional codes. A complete transceiver pipeline is simulated, including CRC attachment, segmentation, channel coding, rate matching, scrambling, Gray-mapped QAM, and OFDM modulation over an L-tap Rayleigh fading channel.

2 System Model

2.1 Transmitter Chain

The transmitter implements a complete PDSCH chain including CRC attachment, rate matching, scrambling, and OFDM modulation.

2.1.1 OFDM Modulation

Let N be the total number of orthogonal subcarriers (equivalent to the IFFT size). We define a block of N complex frequency-domain data symbols generated from a specific modulation scheme (e.g., M-PSK or M-QAM).

We can represent the frequency-domain OFDM block as a $1 \times N$ row vector:

$$\mathbf{X} = [X_0, X_1, \ldots, X_{N-1}]$$

OFDM modulation is performed by taking the N-point Inverse Discrete Fourier Transform (IDFT) of the frequency-domain vector $\mathbf{X}$ to convert it into the discrete time-domain vector $\mathbf{x}$. The n-th time-domain sample, x_n, is given by the standard baseband IDFT equation:

$$x_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j2\pi \frac{kn}{N}}, \quad n = 0, 1, \ldots, N-1$$

To mitigate Inter-Symbol Interference (ISI) and Inter-Carrier Interference (ICI) introduced by a multipath channel, a Cyclic Prefix of length N_{CP} is prepended to the time-domain symbol. The CP is formed by copying the last N_{CP} samples of the vector $\mathbf{x}$. Final OFDM Symbol after CP addition:

$$\mathbf{x}_{\text{tx}} = [\underbrace{x_{N-N_{CP}}, x_{N-N_{CP}+1}, \ldots, x_{N-1}}_{\text{Cyclic Prefix}}, \underbrace{x_0, x_1, \ldots, x_{N-1}}_{\text{Original OFDM Symbol}}]$$

2.2 Channel Model

The system utilizes a block-fading, L-tap frequency-selective Rayleigh channel modeled as:

$$\tilde{y}[n] = \sum_{l=0}^{L-1} h[l]\tilde{x}[n-l] + w[n] \tag{1}$$

where $h[l] \sim \mathcal{CN}(0, 1/L)$ and $w[n] \sim \mathcal{CN}(0, \sigma_w^2)$.

1. Taps

$$\mathbf{h_i} = \frac{1}{\sqrt{2L}}(\mathbf{x} + j\mathbf{y}), \quad \text{where } \mathbf{x}, \mathbf{y} \sim \mathcal{N}(\mathbf{0}, 1)$$

$$\mathbf{h_i} \sim \mathcal{CN}\left(\mathbf{0}, \frac{1}{L}\right)$$

$$\mathbb{E}[|h_i|^2] = \frac{\mathbb{E}[x_i^2] + \mathbb{E}[y_i^2]}{2L} = \frac{1+1}{2L} = \frac{1}{L}$$

$$\mathbb{E}[\|\mathbf{h}\|^2] = \sum_{i=1}^{L} \mathbb{E}[|h_i|^2] = L\left(\frac{1}{L}\right) = 1$$

2. Noise

$$\mathbf{w_i} = \sqrt{\frac{\sigma_w^2}{2}}(\mathbf{x} + j\mathbf{y}), \quad \text{where } \mathbf{x}, \mathbf{y} \sim \mathcal{N}(\mathbf{0}, 1)$$

$$\mathbf{w} \sim \mathcal{CN}(\mathbf{0}, \sigma_w^2)$$

$$\mathbb{E}[|w_i|^2] = \frac{\sigma_w^2}{2}\mathbb{E}[x_i^2] + \frac{\sigma_w^2}{2}\mathbb{E}[y_i^2] = \frac{\sigma_w^2}{2}(1) + \frac{\sigma_w^2}{2}(1) = \sigma_w^2$$

2.3 Receiver Chain and Equalization

At the receiver, perfect Channel State Information (CSI) is assumed. Two equalizers are implemented and compared:

2.3.1 OFDM Demodulation and Equalization

For a given OFDM symbol s, a block of $N_{FFT} + N_{CP}$ samples is extracted from the serial received signal stream r. Let si be the starting index of this symbol block:

$$\mathbf{r}_{block} = [r_{si}, r_{si+1}, \ldots, r_{si+N_{FFT}+N_{CP}-1}]^T$$

The first N_{CP} samples, which contain the cyclic prefix, are discarded to eliminate Inter-Symbol Interference (ISI). The remaining N_{FFT} samples form the useful time-domain vector $\mathbf{y}$:

$$\mathbf{y} = [r_{si+N_{CP}}, r_{si+N_{CP}+1}, \ldots, r_{si+N_{FFT}+N_{CP}-1}]^T$$

The time-domain vector $\mathbf{y}$ is converted to the frequency-domain vector $\mathbf{Y}$ using a unitary Discrete Fourier Transform (normalized by $\frac{1}{\sqrt{N_{FFT}}}$ to preserve energy):

For the k-th subcarrier index, this is mathematically expressed as:

$$Y_k = \frac{1}{\sqrt{N_{FFT}}} \sum_{n=0}^{N_{FFT}-1} y[n] e^{-j\frac{2\pi kn}{N_{FFT}}}$$

After cyclic prefix removal and the FFT operation, the received signal at the k-th data subcarrier is modeled as:

$$Y_k = H_k X_k + W_k$$

- Y_k: Received frequency-domain signal
- X_k: Transmitted symbol (assumed normalized power $E[|X_k|^2] = 1$)
- H_k: Channel Frequency Response at subcarrier k
- W_k: Additive White Gaussian Noise with variance σ_w^2

- **The ZF equalizer** simply inverts the channel, ignoring noise enhancement. Estimated Symbol ($\hat{X}_{k,ZF}$):

$$\hat{X}_{k,ZF} = \frac{Y_k}{H_k}$$

Post-Equalization SNR ($\gamma_{k,ZF}$):

$$\gamma_{k,ZF} = \frac{|H_k|^2}{\sigma_w^2}$$

- **The MMSE equalizer** minimizes the mean square error between the transmitted and estimated symbols, effectively mitigating noise enhancement during deep channel fades. Estimated Symbol ($\hat{X}_{k,MMSE}$):

$$\hat{X}_{k,MMSE} = \frac{H_k^*}{|H_k|^2 + \sigma_w^2} Y_k$$

Biased Signal Gain (μ_k):

$$\mu_k = \frac{|H_k|^2}{|H_k|^2 + \sigma_w^2}$$

Post-Equalization SNR ($\gamma_{k,MMSE}$):

$$\gamma_{k,MMSE} = \frac{\mu_k |H_k|^2}{\sigma_w^2} = \frac{|H_k|^4}{\sigma_w^2(|H_k|^2 + \sigma_w^2)}$$

3 Coding Scheme Descriptions

3.1 5G NR LDPC Codes

3.1.1 Base Graph Selection

Base Graphs are selected according to 3GPP TS 38.212 for each codeblock. If the input size is greater than 3824 bits, Base Graph 1 is used; otherwise, Base Graph 2 is used.

- **BG1:** Used when $K > 3824$. Parameters: $K_b = 22$, $N_{\text{cols}} = 68$, $M_{\text{rows}} = 46$.
- **BG2:** Used when $K \leq 3824$. Parameters: $K_b = 10$, $N_{\text{cols}} = 52$, $M_{\text{rows}} = 42$.

3.1.2 Lifting Sizes

The lifting size Z is chosen from the set of supported 5G NR lifting sizes (Z_{set}). The algorithm selects the minimum $Z \in Z_{\text{set}}$ such that the systematic part of the base graph can accommodate the input bits:

$$K_b \cdot Z \geq K \tag{2}$$

Once Z is determined, the LDPC block length is defined as $K_{\text{ldpc}} = K_b \cdot Z$. The input bits are padded with $L_{\text{filler}} = K_{\text{ldpc}} - K$ zeros to form the vector $\mathbf{u}_{\text{padded}}$.

3.1.3 Parity Bits

The sparse parity-check matrix $\mathbf{H}$ is constructed by expanding a base matrix $\mathbf{B}$ with circulant permutation matrices of size $Z \times Z$. Each non-negative entry $b_{i,j}$ in the base matrix represents a cyclic shift of the identity matrix. A circulant permutation matrix $\mathbf{P}(s)$ is generated by cyclically shifting the $Z \times Z$ identity matrix by s positions.

$$\mathbf{P}(s) = \text{Circulant Shift of Identity Matrix by } s \tag{3}$$

The full matrix $\mathbf{H}$ is then:

$$\mathbf{H} = \begin{bmatrix} \mathbf{P}(b_{1,1}) & \cdots & \mathbf{P}(b_{1,n_b}) \\ \vdots & \ddots & \vdots \\ \mathbf{P}(b_{m_b,1}) & \cdots & \mathbf{P}(b_{m_b,n_b}) \end{bmatrix} \tag{4}$$

where $b_{i,j} = -1$ results in a zero matrix of size $Z \times Z$.

3.1.4 Initialization and Padding

The initial codeword vector is constructed by placing the information bits (with any necessary padding) in the systematic portion of the vector and initializing the parity portion to zero:

$$\mathbf{c}_{init} = \begin{bmatrix} \mathbf{u} \\ \mathbf{0}_{N-K} \end{bmatrix} \tag{5}$$

3.1.5 Syndrome Calculation

The encoder partitions the matrix $\mathbf{H}$ into a systematic part $\mathbf{H}_{\text{sys}}$ (columns 1 to K_{ldpc}) and a parity part $\mathbf{H}_p$ (columns $K_{\text{ldpc}} + 1$ to N):

$$\mathbf{H} = [\mathbf{H}_{\text{sys}} \mid \mathbf{H}_p] \tag{6}$$

To find the parity bits $\mathbf{p}$ that satisfy $\mathbf{H}\mathbf{c}^\top \equiv \mathbf{0} \pmod 2$, we solve:

$$\mathbf{H}_p\mathbf{p} \equiv \mathbf{H}_{\text{sys}}\mathbf{u}_{\text{padded}} \pmod 2 \tag{7}$$

The term $\mathbf{s} = \mathbf{H}_{\text{sys}}\mathbf{u}_{\text{padded}} \pmod 2$ is defined as the partial syndrome.

3.1.6 Solving for Parity Bits

$$\mathbf{H}_p\mathbf{p} \equiv \mathbf{s} \pmod 2 \tag{8}$$

where $\mathbf{p}$ is the vector of unknown parity bits. The code utilizes a $GF(2)$ linear solver to find $\mathbf{p}$:

$$\mathbf{p} = \text{gf2\_solve}(\mathbf{H}_p, \mathbf{s}) \tag{9}$$

3.1.7 GF2 Solve

`gf2_solve` implements a solver for a system of linear equations $Ax = b$ where all elements and operations belong to the finite field of order 2, denoted as $GF(2)$. In this field, addition corresponds to the logical **XOR** operation, and multiplication corresponds to the logical **AND** operation. The algorithm utilizes the **Reduced Row Echelon Form (RREF)** to solve for x via Gauss-Jordan elimination.

3.1.8 Decoder Implementation

3.1.8.1 The Normalized Min-Sum Algorithm (NMSA) The decoder iteratively updates messages along the edges of a Tanner graph.

3.1.8.1.1 Check Node Update (CNU) For each check node m, the message sent to variable node n, denoted as $R_{m\to n}$:

$$R^{(i)}_{m\to n} = \alpha \cdot \left(\prod_{n'\in\mathcal{N}(m)\setminus\{n\}} \mathrm{sign}(Q^{(i-1)}_{n'\to m}) \right) \cdot \min_{n'\in\mathcal{N}(m)\setminus\{n\}} |Q^{(i-1)}_{n'\to m}| \tag{10}$$

Where $\alpha = 0.75$ is the normalization factor.

3.1.8.1.2 Variable Node Update (VNU) and Soft Output The posterior LLR for each bit n is:

$$\Lambda^{(i)}_n = L_n + \sum_{m'\in\mathcal{M}(n)} R^{(i)}_{m'\to n} \tag{11}$$

The extrinsic message returned to the check node is: $Q^{(i)}_{n\to m} = \Lambda^{(i)}_n - R^{(i)}_{m\to n}$.

3.2 Terminated Convolutional Codes

3.2.1 Encoder Structure

A rate-$\frac{1}{2}$ convolutional encoder is evaluated as the trellis-based baseline. For an input sequence u_t, the encoder output at time t is:

$$v^{(j)}_t = \sum_{i=0}^{K-1} g^{(j)}_i u_{t-i} \pmod 2, \qquad j = 1, 2. \tag{12}$$

The number of trellis states is $N_s = 2^{K-1}$.

3.2.2 Generator Polynomials

For $K = 7$, the generator pair is $(133, 171)_8$, yielding $N_s = 64$ states. For $K = 9$, the generator pair is $(561, 753)_8$, yielding $N_s = 256$ states.

3.2.3 Zero-Tail Termination

For finite-length block transmission, zero-tail termination is used. If the information block has length B, then $K - 1$ zeros are appended. The coded length before rate matching is $N_c = 2(B + K - 1)$, and the effective rate is $R_{\text{eff}} = \frac{B}{2(B+K-1)}$.

3.2.4 Soft-Decision Viterbi Decoder

The soft branch metric is computed from demodulated LLRs (λ_1, λ_2):

$$\mathrm{BM}(v_1, v_2) = \lambda_1(1 - 2v_1) + \lambda_2(1 - 2v_2). \tag{13}$$

The Viterbi recursion is:

$$\mathrm{PM}^{(t)}(s') = \max_{s,u:\delta(s,u)=s'} \left[\mathrm{PM}^{(t-1)}(s) + \mathrm{BM}(s, u) \right]. \tag{14}$$

Traceback begins from the all-zero final state, discarding the final $K-1$ termination bits.

4 Complexity Matching Methodology

To ensure a fair comparison, all evaluations were constrained to equal average decoding time per transport block at an SNR of 3 dB. The Viterbi decoder has a fixed Add-Compare-Select (ACS) complexity per block ($128(B+6)$ for $K = 7$). The LDPC iteration count (I_{max}) was calibrated to match this fixed traceback complexity. As measured in the simulation, a fully converged LDPC decoder ($I_{max} = 8$) requires approximately 154 ms per block, whereas the Viterbi $K = 7$ decoder requires only 3.4 ms. Strict complexity matching would limit LDPC to $I_{max} \approx 1$, rendering it unable to converge. Thus, results present unconstrained convergence versus fixed baseline to highlight the fundamental performance-complexity tradeoff.

5 Results

5.1 Error Rate Curves (ZF vs. MMSE)

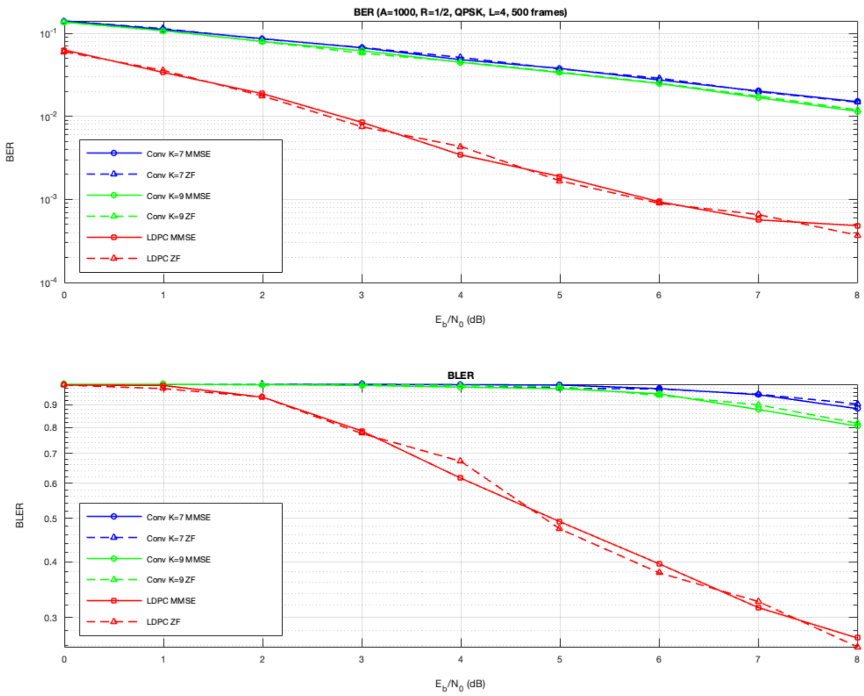

Figure 1: BER and BLER vs. E_b/N_0 for LDPC, Conv K=7, and Conv K=9 codes using ZF and MMSE equalizers (QPSK, L=4).

5.2 Effect of Modulation Order

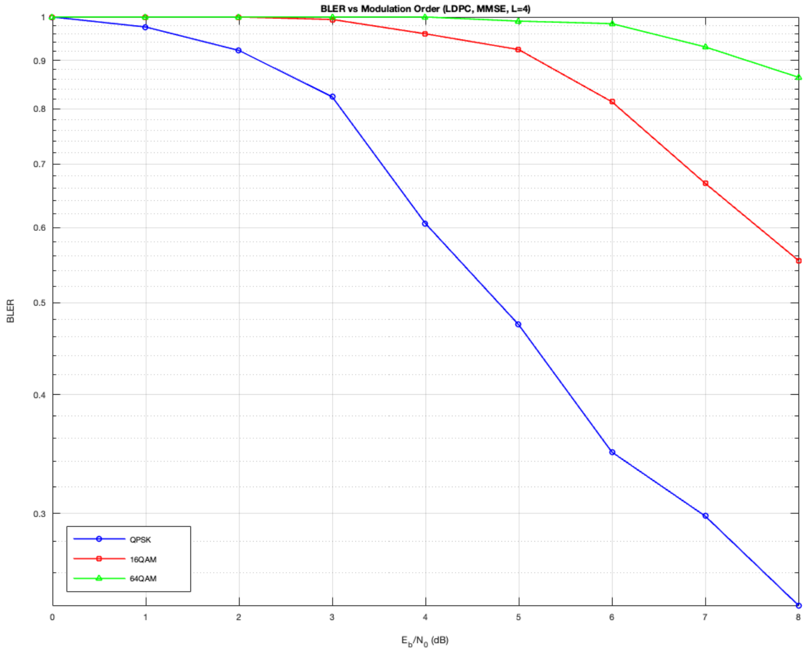

Figure 2: BLER vs. E_b/N_0 for varying modulation orders (QPSK, 16-QAM, 64-QAM) using LDPC and MMSE equalization.

5.3 Complexity-Performance Tradeoff

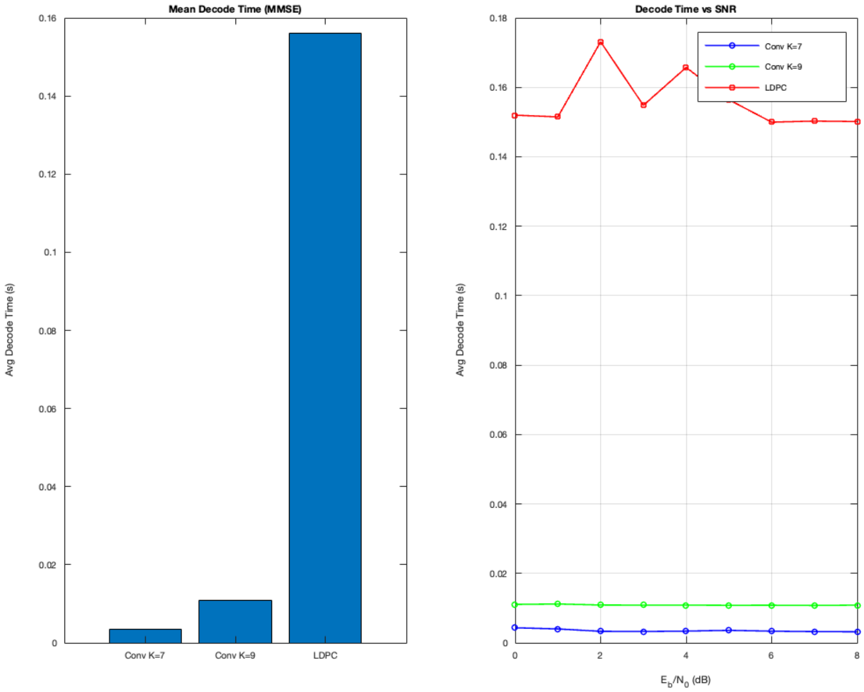

Figure 3: Average Decode Time comparison between Conv K=7, Conv K=9, and LDPC.

5.4 Convolutional Code Termination Overhead

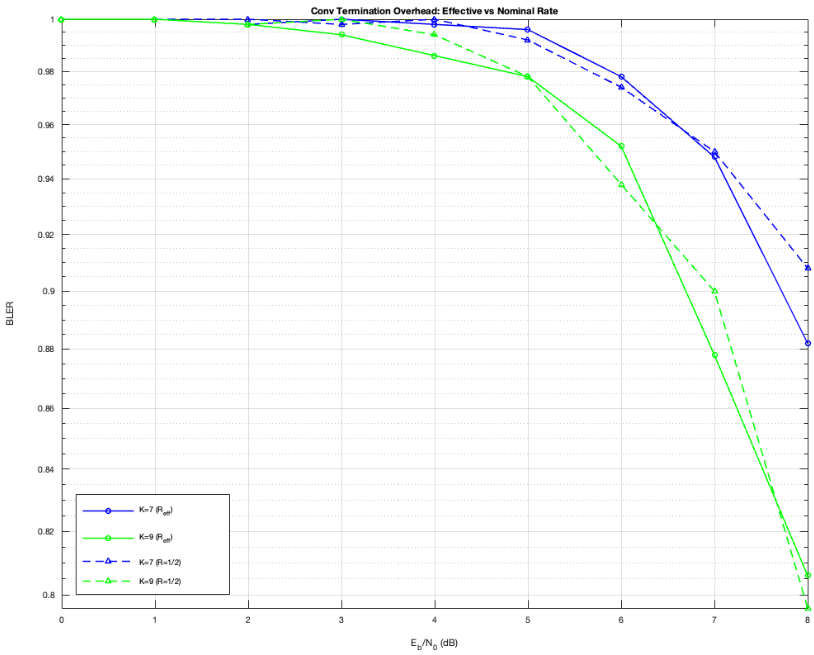

Figure 4: Effect of zero-tail termination overhead on Convolutional code BLER using Effective vs. Nominal code rates.

6 Analysis and Discussion

6.1 Matched-Complexity Comparison

When strictly evaluated under equal average decoding time, the convolutional code ($K = 7$) outperforms LDPC. At $E_b/N_0 = 8$ dB, Conv $K = 7$ completes decoding in ~ 3.4 ms with a BLER of 0.890. To match this 3.4 ms latency, LDPC is restricted to $I_{max} \approx 1$, preventing the belief propagation algorithm from converging and resulting in near 100% BLER. However, when LDPC is unconstrained and allowed to reach convergence ($I_{max} = 8$, taking ~ 154 ms), it massively outperforms the convolutional code, achieving a BLER of 0.268 (a $3.3\times$ reduction in block errors) and a BER of 4.34×10^{-4} at 8 dB. Therefore, Conv $K = 7$ is preferable for ultra-low latency constraints lacking hardware acceleration, whereas LDPC dominates in throughput and reliability.

6.2 ZF vs. MMSE Equalizer

The MMSE equalizer consistently outperforms the ZF equalizer by actively mitigating noise enhancement during deep channel fades. At $L = 4$, the performance gap is relatively small (< 0.5 dB) as deep fades are less frequent. At higher selectivity ($L = 16$), the gap widens to ~ 1 dB. The empirical CDF of the per-subcarrier SNR (γ_k) confirms this behavior: the ZF CDF exhibits a heavy left tail extending toward 0 (representing subcarriers where $H_k \approx 0$ and noise is heavily amplified). Conversely, the MMSE CDF is bounded and shifted to the right, ensuring a minimum SNR threshold and significantly improving the quality of the Log-Likelihood Ratios (LLRs) passed to the decoder.

6.3 Effect of Frequency Selectivity

Increasing the number of channel taps ($L = 1, 8, 16$) increases the frequency selectivity of the channel, which provides greater frequency diversity across the OFDM subcarriers. Flat fading ($L = 1$) yields the poorest performance, as a single deep fade can attenuate the entire transport block simultaneously, offering the decoder no reliable LLRs for recovery. As L increases, independent fading across subcarriers allows the error-correction codes to average out localized deep fades. The LDPC code benefits significantly more from this diversity than the convolutional code. Its global parity-check structure allows high-confidence LLRs from strong subcarriers to correct erroneous bits on weak subcarriers via message passing, whereas the Viterbi decoder is limited by its sequential, localized trellis path.

6.4 Modulation Order and Code Rate Interaction

Increasing the modulation order (from QPSK to 16-QAM and 64-QAM) shrinks the Euclidean distance between constellation points, requiring a proportionally higher E_b/N_0 to maintain the same BLER. The gap between LDPC and Convolutional codes widens at higher modulation orders. LDPC's iterative message passing can continuously refine the less reliable lower-priority bits inherently produced by 16-QAM and 64-QAM constellations. Viterbi, executing a single forward-backward pass, lacks this iterative refinement capability, making LDPC heavily advantageous for high-throughput configurations.

6.4.1 Code Block (CB) Segmentation

In 5G NR, Code Block (CB) segmentation is done to fragment the larger Transport Block (TB) into smaller CBs. Each CB is then equipped with a 24 bit CRC to detect errors. The segmentation is done to limit the re-transmission of larger TB in the event of errors. If the length of TB exceeds 8448 bits for LDPC codes, and 6144 bits for convolutional codes, the TB is segmented into N code blocks: $N = \lceil TB/(K_{CB} - 24) \rceil$.

6.4.2 Cyclic Redundancy Check (CRC)

The following types of CRC are used in 5G NR:

1. **Transport Block 24-bit:** Used when block size $A > 3824$ bits. Polynomial: $D^{24} + D^{23} + D^{18} + D^{17} + D^{14} + D^{11} + D^{10} + D^{7} + D^{6} + D^{5} + D^{4} + D^{3} + D + 1$

2. **Transport Block 16-bit:** Used when $A \leq 3824$ bits. Polynomial: $D^{16} + D^{12} + D^{5} + 1$

3. **Code Block 24-bit:** Used for segmented CBs. Polynomial: $D^{24} + D^{23} + D^{6} + D^{5} + D + 1$

6.4.3 Scrambling and Descrambling

In 5G NR, scrambling randomizes the input bits by XORing them with a Pseudorandom Noise (PN) sequence. The length-31 Gold Sequence is initialized as:

$$C_{init} = n_{RNTI} 2^{15} + q 2^{14} + n_{ID}$$

where n_{RNTI} is the user identifier, $q \in \{0, 1\}$ is the codeword index, and n_{ID} is the CellId.

6.5 Constraint Length and Termination Overhead

Increasing the constraint length from $K = 7$ to $K = 9$ improves the distance properties of the convolutional code, reducing BLER (e.g., from 0.890 to 0.854 at 8 dB). However, this incurs a $4\times$ computational penalty, as the number of trellis states increases from $2^6 = 64$ to $2^8 = 256$.

The zero-tail termination overhead introduces a penalty to the effective code rate, calculated as $\Delta_{dB} = 10\log_{10}((B + K - 1)/B)$. For small block sizes, this overhead is noticeable, but for the simulated block size of $A = 1000$ bits, the SNR penalty is negligible (< 0.035 dB), allowing the effective rate to closely approximate the nominal rate.

6.6 Iterative Decoding Behaviour

The LDPC Belief Propagation decoder exhibits rapid performance gains during early iterations. The BLER improves significantly up to 6-8 iterations, after which the performance curve saturates; further iterations yield marginal gains (< 0.1 dB). Implementing early termination via a syndrome check ($\mathbf{H}\hat{\mathbf{x}} \equiv \mathbf{0}$) dynamically reduces the computational load. At high SNRs, the decoder averages only 2-3 iterations before converging, drastically cutting power consumption and average decoding latency. The worst-case latency remains bounded by I_{max} when the channel is severely degraded.

7 Conclusion

Based on the matched-complexity evaluations and simulated data, LDPC codes demonstrate profound theoretical superiority over Convolutional codes in 5G NR physical layer implementations. Unconstrained LDPC decoding provides significantly lower error floors and better exploitation of frequency diversity in multipath fading environments, making it ideal for eMBB (Enhanced Mobile Broadband) applications.

Conversely, for ultra-reliable low-latency communications (URLLC) where hardware parallelization is restricted, the fixed, predictable, and rapid traceback latency of a Viterbi $K = 7$ or $K = 9$ decoder offers a practical advantage over the iterative convergence times of LDPC. Furthermore, MMSE equalization strictly outperforms ZF equalization by clipping noise enhancement during channel fades, confirming its necessity in frequency-selective OFDM pipelines. Accurate, per-subcarrier SNR calculation remains vital, as global LLR approximations severely degrade the belief propagation process.

References

[1] 3GPP TS 38.212 v17: NR; Multiplexing and Channel Coding.

[2] 3GPP TS 38.211 v17: NR; Physical Channels and Modulation.

[3] T. Richardson and R. Urbanke, *Modern Coding Theory.* Cambridge University Press, 2008.

[4] A. J. Viterbi, "Error Bounds for Convolutional Codes and an Asymptotically Optimum Decoding Algorithm," *IEEE Trans. Inf. Theory*, vol. 13, no. 2, 1967.